PARAMPREET SINGH

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PROFESSIONAL SUMMARY

Final-year Computer Science undergraduate (CGPA 9.44) with a strong foundation in data structures, algorithms (350+ problems), and system design. Full-stack software developer skilled in Java, Python, and TypeScript, building scalable web applications, REST and real-time APIs, and cloud-deployed systems on AWS. Fast learner who takes ownership from requirements to delivery. Seeking Software Developer roles.

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, TypeScript, SQL, Bash
CS Fundamentals: Data Structures and Algorithms (350+ problems), System Analysis and Design, OOP, OS, DBMS, Computer Networks
UI Development: React.js, Next.js (App Router), TypeScript, HTML, CSS, Tailwind CSS, Zustand, React Query, Responsive Design
Backend and APIs: Node.js, Express.js, RESTful APIs, XML/HTTP Integrations, JWT, OAuth, RBAC, LLM/OpenAI API Integration
Real-Time and Async: WebSockets, WebRTC, Redis, BullMQ Job Queues, Scheduled Jobs, n8n Automation
Databases: PostgreSQL, MySQL, MongoDB - schema design, ORMs (Prisma, Drizzle, Mongoose)
Cloud and Tools: AWS (EC2, S3, RDS, ECS, ECR, SNS, CloudWatch), Docker, Kubernetes, Terraform, Nginx, Git, GitHub Actions (CI/CD)

EDUCATION

Bachelor of Technology in Computer Science
Chitkara University, Rajpura, Punjab | CGPA: 9.44 | Final Year

EXPERIENCE

Software Developer Intern | Rabbitt AI | April 2026 - Present

Tally Reconciliation and Accounting Automation Platform | Client Project
Next.js, React, Node.js, TypeScript, PostgreSQL (AWS RDS), Prisma, Redis, BullMQ, Docker Compose, Nginx, AWS EC2, TallyPrime XML/HTTP

- Fully managed the project end to end: gathered requirements, led client calls and product demos, translated accounting rules into technical design, and delivered the platform for 45+ accounting users.
- Architected a Next.js frontend, Node.js/TypeScript REST backend, and separate BullMQ worker service with Redis and PostgreSQL on AWS RDS, containerized with Docker Compose behind an Nginx reverse proxy on AWS EC2.
- Built ingestion pipelines for seven invoice REST APIs, payment-gateway and bank reports, and email/Excel imports, with per-source configurable field mappings, schema normalization, and source-level deduplication.
- Implemented a strict one-to-one reconciliation engine matching on transaction date, category, reference, and exact amount, classifying mismatches, one-sided records, and duplicates into an exception-review workflow.
- Generated TallyPrime XML vouchers with ledger, Cost Centre, and GST/HSN mappings, gated by a server-side preflight that verifies the Tally endpoint and locks the target company by GUID before posting.
- Engineered fault-tolerant batch posting for up to 13,000 vouchers with per-company concurrency control, persistent idempotency keys, verify-before-retry on timeouts, a resumable Pending state, and a scheduled daily pipeline with full audit logs.

DRIP | Full-Stack Coding Assessment Platform
Node.js, Express.js, Next.js, React, TypeScript, MongoDB, Mongoose, WebSockets, Docker, AWS ECS, S3, OpenAI API, Zustand

- Engineered a coding-assessment platform where users launch isolated browser-based IDE sessions, run test suites, submit solutions, and receive automated grading with AI-assisted feedback via the OpenAI API.
- Designed a session-orchestration service that provisions per-user Docker containers on AWS ECS for Python/ML, cloud, and React/Node workloads, tracking session and attempt lifecycle in MongoDB.
- Built a WebSocket proxy layer routing real-time, bidirectional traffic between the browser and container-hosted code-server instances.
- Developed modular RESTful APIs with Express.js and Mongoose, secured by JWT authentication and role-based access control across students, administrators, classrooms, assessments, sessions, and attempts.
- Built a Next.js App Router frontend in TypeScript with centralized Zustand stores, and offloaded file delivery to Amazon S3 presigned URLs to reduce backend load.

CloudWatcher | Cloud Observability Tool
Node.js, Python, AWS SDK, AWS CloudWatch, SNS, IAM, Terraform, GitHub

- Built Node.js and Python services that pull AWS CloudWatch metrics, evaluate threshold rules, and publish automated alerts through Amazon SNS.
- Authored reusable, version-controlled Terraform modules provisioning CloudWatch alarms, SNS topics, and IAM roles, shipped through feature branches, pull-request code reviews, and issue-based contributions.

PROJECTS

CampusBarter | Full-Stack Web Application | Live App
https://campusbarter.onrender.com
React, Node.js, Express, PostgreSQL, Drizzle ORM, WebSockets, Passport.js, Tailwind CSS, React Query, Zod

- Built a peer-to-peer campus marketplace with Passport.js session authentication (Google OAuth and local login) and real-time chat over WebSockets.
- Designed a relational schema on serverless PostgreSQL with Drizzle ORM for type-safe queries; used React Query for server-state caching and Zod for input validation; deployed on Render.

TisiOps | AI-Powered DevOps Platform | Founder and Developer | GitHub
https://github.com/ParampreetSingh23/TisiOps
Next.js, TypeScript, Node.js, PostgreSQL, Prisma, Redis, BullMQ, Docker, Terraform, AWS EC2, GitHub API, Clerk, OpenTelemetry

- Built a platform where specialised AI agents plan cloud provisioning, deployment, staging, monitoring, and infrastructure diagnosis, executed through a Plan, Validate, Approve, Execute, Verify pipeline instead of direct LLM shell access.
- Developed production-to-staging automation that analyses GitHub repos and live servers, detects runtime, services, and dependencies, generates isolated staging plans, provisions AWS EC2 with Terraform, and deploys with Docker.
- Implemented asynchronous execution on Redis/BullMQ with retries, idempotency, job tracking, SSH readiness checks, and PostgreSQL-persisted state; added OpenTelemetry/SigNoz monitoring and Clerk user-scoped security.

CERTIFICATIONS

- IBM - Introduction to Agile Development and Scrum
- IBM - Introduction to DevOps